

Notes on Isolation Forest: a geometric point of view

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Abstract

These notes present a simple framework for visualizing the anomaly score function of an Isolation Forest. The proposed approach can also be extended to other models that provide a numerical score function.

1 A visualization approach

Anomaly detection is used in applications such as fraud detection, network security, and system monitoring. However, interpreting how an anomaly score behaves across a high-dimensional feature space is not straightforward. These notes present a simple visualization framework for the anomaly score function which can also be extended to other models that provide a numerical score function. The key idea is to visualize a two-dimensional slice of the fitted score function while keeping the remaining features fixed at specified values:

- **feature selection:** let $\mathbf{X} \in \mathbb{R}^{n \times d}$ denote the dataset, where n is the number of observations and d is the number of features. We define a variability measure $V(X_j)$ for each feature X_j , $j = 1, \dots, d$. The two selected features are those corresponding to the two largest values of V :

$$V(X_{j_1}) \geq V(X_{j_2}) \geq V(X_j), \quad \forall j \notin \{j_1, j_2\}.$$

For example, $V(X_j)$ may be defined as the second central moment of the empirical distribution:

$$V(X_j) = \frac{1}{n} \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2, \quad \bar{x}_j = \frac{1}{n} \sum_{i=1}^n x_{ij}.$$

The selected features are denoted by X_{j_1} and X_{j_2} . This selection is used only to define the two-dimensional representation of the observations in the subsequent visualization. It is not intended to identify the features that are most important for the Isolation Forest or to provide a model-based feature-importance measure.

- **data preprocessing and model training:** before the fitting, the input data are transformed or standardized according to the adopted preprocessing procedure. Let $\mathbf{X}^* \in \mathbb{R}^{n \times d}$ denote the resulting dataset. The Isolation Forest is then trained on $\mathbf{X}^*$ using the adopted model and hyperparameter configuration.
- **anomaly score:** after training, the fitted Isolation Forest is used to compute a score for each observation. Let $\mathbf{x}_i^* \in \mathbb{R}^d$ denote the i -th preprocessed observation. Its corresponding model score is $s_i = s(\mathbf{x}_i^*)$ for $i = 1, \dots, n$ where $s : \mathbb{R}^d \rightarrow \mathbb{R}$ denotes the score function of the trained Isolation Forest. For example, in Python, the `score_samples` function using `scikit-learn` returns scores for which lower values correspond to observations considered more anomalous. For visualization purposes, we define the transformed anomaly score $a(x) = -s(x)$ where $a_i = a(x_i^*) = -s_i$. So larger values of a_i indicate observations considered more anomalous by the model.
- **visualization:** the two selected features X_{j_1} and X_{j_2} define the horizontal and vertical coordinates while the anomaly score determines the color. Scores are computed from the full preprocessed vectors $\mathbf{x}_i^*$, allowing us to visualize the observations and their corresponding model scores in two dimensions.

1.1 A numerical example

We consider a synthetic dataset with two dominant features and a small fraction of anomalous observations. Normal observations are generated from a Gaussian distribution and anomalous ones are generated randomly in the feature space. The Isolation Forest is trained on standardized data using `scikit-learn`. We visualize the observations using the two selected features and evaluate the fitted score function over a regular grid, keeping the remaining features fixed at specified representative values. The resulting heatmap and three-dimensional surface therefore represent a two-dimensional slice of the original d -dimensional score function. The score sign is reversed so that higher values indicate more anomalous regions, allowing us to inspect how the fitted anomaly score varies across the selected subspace.

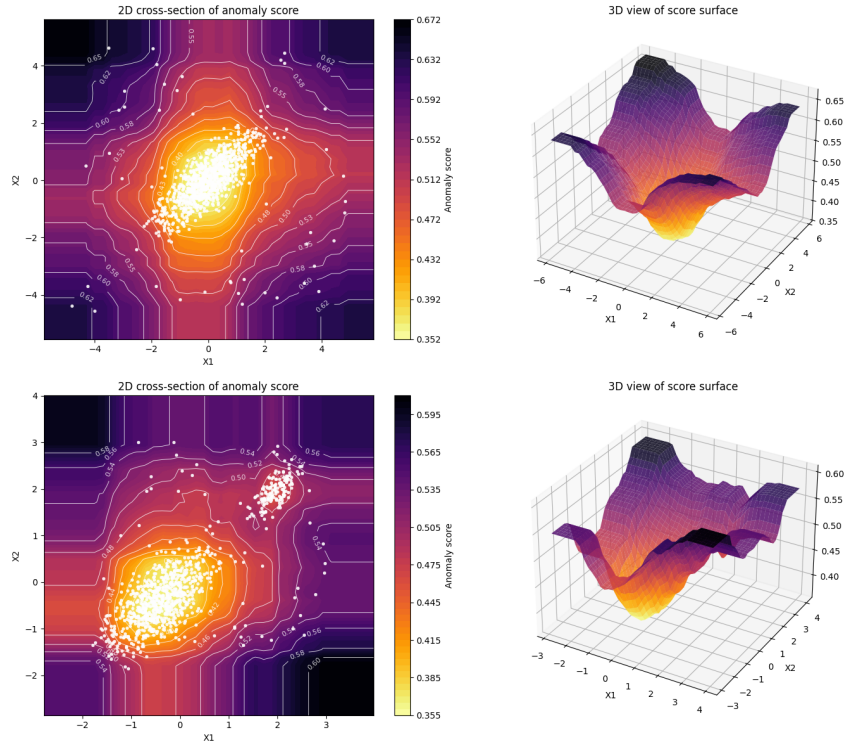


Figure 1: In this example, the visualization suggests that observations located in relatively sparse regions of the feature space tend to lie in regions associated with higher anomaly scores. The surface represents a two-dimensional slice of the fitted score function, obtained by fixing the remaining features at specified values.

2 Limitations

The proposed approach is intentionally simple and should be regarded as an exploratory visualization rather than a complete interpretability method. Some limitations should be kept in mind:

- **exploratory nature:** the visualization can provide useful intuition and suggest potentially relevant patterns or interactions but these observations should be further investigated using more formal feature-analysis methods;
- **feature selection:** the two features do not necessarily need to be selected according to a specific statistical criterion. They can also be chosen manually based on the particular question or application of interest. For example, one may choose to visualize two features because their relationship is of practical interest;
- **two-dimensional representation:** the visualization only shows the score function with respect to two features at a time. It therefore does not represent the complete score function in the original high-dimensional feature space, and patterns involving other features or higher-order interactions may not be visible;
- **features conditioning:** the two-dimensional score landscape depends on the values assigned to the remaining features. Fixing these features at mean, median, or other representative values produces only one slice of the original score function. Different choices of the fixed values may therefore lead to different visualizations.

Conclusion

These notes present a simple way to visualize the behavior of an Isolation Forest. By selecting two features and varying them while keeping the remaining features fixed, the approach provides a two-dimensional view of the anomaly score function. Although it does not capture the full high-dimensional score landscape, the visualization can help reveal patterns or regions of interest that may be difficult to identify from the numerical scores alone. These patterns can then be further investigated using more formal analysis or interpretability methods.